

A Child Document Under the DBA-MA International Framework

DBA-MA L4 — IAEA Monitoring Guidance

*Monitoring Sources and Data-Sharing Guidance for DBA-MA
Trigger Observables*

Version 3.0

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1. Purpose and Scope

The DBA-MA Trigger Framework (DBA-MA L3) defines observable indicators that govern the transition of a nuclear facility from normal operations through preparation for a controlled shutdown to the achievement of the Autonomous Safe Shutdown Condition (ASSC). For those indicators to function as intended, the facility must actively monitor the right sources and have established the information-sharing relationships necessary to receive external threat data before a crisis develops.

This document provides implementation guidance for asset owner/operators (AOOs) on where to monitor each Tier 1 observable indicator, which internal and external sources are relevant, which data-sharing agreements are required, and how commercial providers can supplement government sources. The objective is to ensure that, when the DBA-MA Trigger Framework is activated, the information pathways are already established and exercised, rather than created for the first time under crisis conditions.

This document establishes a jurisdiction-neutral framework for monitoring source categories. The specific organizations, programs, and agreements applicable in each national implementation are addressed in the corresponding national child document. DBA-MA L4, United States National Implementation, sets out the United States implementation of the monitoring framework; the Finnish implementation is addressed in the Finnish national child document (DBA-MA L1, Finland).

This guidance addresses the Tier 1 (Strategic Warning) observables. Tier 2 and Tier 3 observables largely depend on the same information pathways established for Tier 1, with the addition of classified military situational awareness that will be addressed through the proposed Nuclear Facility Military Threat Coordination Protocol described in the applicable national implementation. Establishing the Tier 1 information infrastructure is a necessary precondition for receiving Tier 2 and Tier 3 information when the threat escalates. The drone observables that DBA-MA L3 carries at each tier depend, in addition, on the facility's own capability to detect and characterize uncrewed aerial systems, addressed in DBA-UxS, Section 4.3, and in DBA-MA-UxS-NA; the review cycle for the drone threat characterization is addressed in Section 5.5 of this document.

2. Observable Monitoring Sources: Summary

The following table maps each Tier 1 observable indicator to the generic categories of monitoring sources, required agreements, and available commercial supplements applicable across all national implementations. Detailed guidance for each observable follows in Section 3. National-specific program names and agreement details are addressed in the applicable national child documents.

Observable	Internal Monitoring	External Government Sources	Required Agreements	Commercial Providers
OBS-1: Increased nation-state cyber reconnaissance against energy sector targets	SOC/NOC telemetry IDS/IPS alerts on OT networks anomalous scanning of SCADA/DCS systems	National electricity sector ISAC alerts national cybersecurity authority notifications National Regulatory Authority Cyber Threat Notifications	Electricity sector ISAC membership National Cybersecurity Information-Sharing Agreement regulatory authority cybersecurity reporting relationship	Managed detection and response providers with OT capability threat intelligence platforms (e.g., Dragos, Claroty, Mandiant) dark web monitoring services
OBS-2: Significant change in volume of state-sponsored cyber operations reporting worldwide	Internal correlation of ISAC and national cybersecurity authority reporting volume trends	National electricity sector ISAC situation reports National Cybersecurity Authority Trend Analysis Allied Nation CERT advisories	Electricity sector ISAC membership automated indicator sharing participation International CERT information sharing	Threat intelligence vendors with global visibility (e.g., Recorded Future, Mandiant) OSINT aggregation services tracking state-sponsored activity
OBS-3: Escalating political or military tensions with a state-level adversary	No direct internal monitoring capability internal analysis function (if staffed), tracking open-source indicators	Foreign ministry public statements and travel advisories National Defense Public Posture Statements NATO and allied public communications	No formal agreement required for open-source monitoring optional classified briefing access through applicable security channels	Geopolitical risk analysis services (e.g., Stratfor/RANE, EIU, Oxford Analytica) specialized conflict and political risk monitoring platforms
OBS-4: National defense condition changes or equivalent national alert status	No direct internal monitoring capability public announcements at higher alert levels classified notification for lower levels	National defense authority notification through the regulatory authority or national security channels national terrorism or infrastructure alert system bulletins	Protected critical infrastructure information program (national equivalent) regulatory authority emergency notification agreements Proposed Nuclear Facility Military Threat Coordination Protocol	Not applicable — classified defense posture information is exclusively government-sourced
OBS-5: Intelligence community assessments of heightened likelihood of military action	No direct internal monitoring capability dependent entirely on external information sharing	Joint intelligence bulletins from national security agencies National Cybersecurity Authority advisories with intelligence-derived content regulatory authority-brokered classified briefings national energy sector risk	Facility security clearances (applicable national level) for key decision-makers classified briefing and information-sharing agreements Proposed Nuclear Facility Military Threat Coordination Protocol	Not applicable — classified intelligence assessments are exclusively government-sourced

		management agency briefings		
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3. Detailed Guidance by Observable

3.1 OBS-1: Increased Nation-State Cyber Reconnaissance Against Energy Sector Targets

3.1.1 What to Monitor

This observable has both internal and external dimensions. Internally, the facility must monitor its IT and OT networks for indicators of nation-state reconnaissance activity, including scanning, probing, attempts to access industrial control systems, credential harvesting, spear-phishing campaigns targeting facility personnel, and anomalous traffic patterns on networks connected to facility control systems. Externally, the facility must monitor threat reporting from government and industry sources for indicators that energy sector targets in the facility's region or nation are experiencing elevated reconnaissance activity, even if the facility's own networks have not been directly targeted.

3.1.2 Internal Monitoring Sources

The facility's Security Operations Center (SOC) and Network Operations Center (NOC) are the primary internal monitoring points. Intrusion detection and prevention systems (IDS/IPS) deployed on both IT and OT networks should be configured with rulesets that flag known nation-state tactics, techniques, and procedures (TTPs). Security information and event management (SIEM) systems should correlate events across IT and OT boundaries. For facilities with operational technology networks running SCADA, DCS, or plant computer systems, monitoring must extend to the OT domain—IT-only monitoring is insufficient, as the most consequential reconnaissance targets the control systems themselves.

The internal monitoring capability must be sufficient to distinguish nation-state reconnaissance from routine background scanning and criminal activity. This requires threat intelligence integration: the SOC must consume and correlate against indicators of compromise (IOCs) associated with known state-sponsored threat actors targeting the energy sector, including groups tracked by the cybersecurity community under designations such as Sandworm and Volt Typhoon, as well as their associated toolsets.

3.1.3 External Government Sources

Each national implementation will identify the specific government programs and agencies that constitute the external government sources for this observable. The generic categories of required sources are: the national electricity sector information-sharing and analysis organization; the national cybersecurity authority (which publishes threat alerts, advisories, and

indicators of compromise); and the national nuclear regulatory authority (which maintains a cybersecurity reporting and notification relationship with licensees). The specific national programs are addressed in the applicable national child documents.

The regulatory authority's cybersecurity notification framework should be verified as bidirectional in practice—that is, the facility should receive regulatory-authority-originated threat notifications, not only report outbound events.

3.1.4 Commercial Providers

Commercial threat intelligence providers specializing in the energy sector and OT can significantly enhance the facility's ability to detect and attribute nation-state reconnaissance. Providers such as Dragos, Claroty, and Mandiant maintain dedicated intelligence teams focused on ICS/SCADA threats and state-sponsored activity against energy infrastructure. These services can provide enriched threat feeds, attribution analysis, and advisory services that supplement government sources. The AOO should evaluate these services based on their OT visibility, energy sector coverage, and ability to provide actionable intelligence relevant to the DBA-MA trigger assessment.

Dark web monitoring services can provide early warning of adversary interest in the facility or the broader energy sector, including pre-attack reconnaissance, credential sales, and communications referencing energy-sector targets.

3.2 OBS-2: Significant Change in State-Sponsored Cyber Operations Reporting

3.2.1 What to Monitor

This observable requires monitoring the volume and patterns of reporting on state-sponsored cyber operations globally, not just within the facility or even the energy sector. The indicator is a change in the baseline—either a significant increase (indicating heightened adversary activity) or a significant decrease (indicating potential operational security measures preceding a major operation). Monitoring this observable requires access to comprehensive threat reporting and the analytical capability to detect changes in patterns.

3.2.2 Monitoring Approach

This is primarily an external monitoring function. The facility is unlikely to have the global visibility needed to detect changes in the volume of state-sponsored cyber operations independently. The monitoring approach should focus on tracking the frequency and urgency of advisories from the national electricity sector ISAC, the national cybersecurity authority, and allied-nation CERTs; monitoring the national cybersecurity authority's active exploitation catalog update frequency as a proxy for the rate of active exploitation of new vulnerabilities; tracking public reporting from major threat intelligence vendors (Mandiant, CrowdStrike, Microsoft Threat Intelligence, Google Threat Analysis Group) for changes in state-sponsored

activity levels; and monitoring public threat assessments from the national intelligence community and relevant governmental bodies.

The critical analytical challenge is establishing the baseline. The facility (or its corporate parent) should maintain a running assessment of the normal volume and cadence of threat reporting, so that a significant change can be recognized when it occurs. This is analogous to the seismic monitoring function at a nuclear plant—the value is in detecting deviation from the baseline, not in any individual reading.

3.2.3 Key Sources and Agreements

The national electricity sector ISAC, the national cybersecurity authority, and the national energy sector risk management agency are the primary government sources for energy-sector-relevant reporting. Commercial threat intelligence vendors with global visibility—particularly those that track state-sponsored threat groups across multiple sectors and geographies—are valuable for this observable because they can detect changes in patterns that sector-specific sources may miss.

3.3 OBS-3: Escalating Political or Military Tensions

3.3.1 What to Monitor

This observable pertains to the geopolitical environment rather than to technical threat data. It requires monitoring for the escalation of tensions between the host nation and a state-level adversary possessing conventional military capability sufficient to conduct operations against nuclear facilities. The relevant indicators are diplomatic, political, and military: troop movements, diplomatic ruptures, escalation of economic sanctions, military exercises near the border, withdrawal of diplomatic personnel, and public rhetoric indicating a willingness to use military force.

3.3.2 Monitoring Approach

Nuclear facility operators have traditionally not maintained a geopolitical intelligence analysis function, and this observable represents a scope expansion for most AOOs. The monitoring can be approached at two levels.

At the open-source level, this observable can be monitored through publicly available information: foreign ministry travel advisories and public statements; national defense press conferences and posture statements; NATO and allied government communications; major news services with foreign affairs coverage; and relevant governmental hearing transcripts on defense and foreign policy topics. This level of monitoring requires no formal agreements and can be performed by existing facility management or corporate staff. The challenge is not access to information but the analytical judgment required to assess when the level of tension has risen to the point of being DBA-MA relevant.

At the structured analysis level, commercial geopolitical risk analysis services can provide professional assessments of political and military risk, including probability-weighted analysis of

conflict escalation. Services such as Stratfor/RANE, the Economist Intelligence Unit, and specialized defense and security consultancies provide this type of analysis. For utility holding companies operating multiple facilities, this function may be most efficiently provided at the corporate level rather than at each facility.

3.3.3 Bridging the Gap: From Geopolitical Awareness to Trigger Assessment

The key challenge with this observable is translating general geopolitical awareness into a specific trigger assessment for the DBA-MA. The Nuclear Facility Military Threat Coordination Protocol proposed in each national implementation is designed to address this by establishing a channel for the intelligence and military communities to communicate threat assessments relevant to the nuclear facility decision. Until that protocol is established, the AOO should designate a specific individual or function to monitor geopolitical developments and assess their relevance to the DBA-MA trigger criteria, and establish a defined process for escalating that assessment to the decision-makers responsible for Tier activation.

3.4 OBS-4: National Defense Condition Changes

3.4.1 What to Monitor

This observable is specific and binary: it requires notification that the national defense posture has changed, as reflected in changes to the national defense readiness condition or equivalent national alert status. Changes at higher alert levels may be publicly announced; changes at lower alert levels may not be disclosed in real time.

3.4.2 The Information Access Problem

This is the first observable in which the AOO encounters a fundamental limitation on information access. The facility cannot independently monitor for changes in defense condition that are not publicly announced. Notification relies entirely on government communication channels, and its timeliness depends on whether the facility is included on the distribution list for such notifications.

In most national implementations, the most reliable pathway is through the national nuclear regulatory authority, which maintains continuous communication with national security agencies and can relay changes in the national defense posture to licensees. However, this relay function may not be formally structured for DBA-MA trigger purposes in any current national implementation, and its timeliness against a defined standard may not be guaranteed. This gap underscores the importance of the Nuclear Facility Military Threat Coordination Protocol in each national implementation.

3.4.3 Required Agreements and Actions

The AOO should verify that the regulatory authority emergency notification pathway is functional and exercised, and that the facility is positioned to receive regulatory-authority-originated notifications about changes in the national threat posture. Enrollment in the national

protected critical infrastructure information program (where one exists) may provide access to additional threat information. Key personnel in the DBA-MA decision chain should hold security clearances sufficient to receive classified briefings when the threat situation warrants.

3.5 OBS-5: Intelligence Community Assessments of Heightened Likelihood of Military Action

3.5.1 What to Monitor

This observable requires receipt of intelligence community assessments that specifically address the likelihood of military action against the host nation or its critical infrastructure. These assessments are classified, are produced by national intelligence agencies, and are not available through open sources or commercial providers. The facility's ability to receive this information depends entirely on government information-sharing decisions and its access to classified channels.

3.5.2 Current Pathways

The most probable current pathway for this information to reach a nuclear facility operator is through joint intelligence bulletins issued by national security agencies at a classified level. The national cybersecurity authority occasionally publishes advisories with intelligence-derived content at lower classification levels. The national nuclear regulatory authority can also broker classified briefings for licensees on specific threat topics. The national energy sector risk management agency, in its capacity as the sector risk management authority, can facilitate the sharing of sector-relevant intelligence assessments with energy-sector entities.

However, none of these pathways is currently structured in any national implementation to deliver the specific type of assessment needed for DBA-MA trigger activation: a time-sensitive, facility-relevant assessment of the likelihood of military action within a timeframe that allows the facility to act. The current pathways are oriented toward general threat awareness rather than driving a specific operational decision at a defined facility.

3.5.3 Required Actions

The AOO should ensure that all personnel in the DBA-MA Tier activation decision chain hold active security clearances sufficient to receive the relevant assessments under the applicable national security framework. The facility should establish or verify access to a classified information facility for receiving classified briefings. The facility should formally request, through its regulatory authority contact and direct engagement with the applicable national security agencies, inclusion in the distribution of sector-specific classified threat briefings.

Most critically, the AOO should actively support and advocate for establishing the Nuclear Facility Military Threat Coordination Protocol as proposed in the applicable national implementation. This is the only proposed mechanism that would create a formal, exercised, real-time pathway for the type of intelligence assessment this observable requires.

4. Agreement and Relationship Readiness Assessment

The following table provides a structured assessment of the data-sharing agreements and relationships required to support DBA-MA trigger monitoring. The AOO should use this table as a self-assessment tool to identify gaps in its current information-access posture and prioritize actions to close them. The "Current Status" column reflects the typical status across the nuclear industry; individual facilities should update this assessment based on their specific situation. National-specific program names corresponding to each generic category are identified in the applicable national child documents.

Agreement / Relationship	Generic Counterparty	Observables Served	Typical Status	Action Required
Electricity sector ISAC membership and information sharing	National electricity sector ISAC	OBS-1, OBS-2	Likely exists where mandatory for grid-registered entities varies otherwise.	Verify active participation confirm receipt of all relevant products, including restricted-distribution advisories.
National cybersecurity authority information sharing (including automated indicator sharing)	National cybersecurity authority	OBS-1, OBS-2	May exist varies by facility and jurisdiction	Enroll in automated indicator sharing if not already participating designate internal POC for authority communications.
National security agency critical infrastructure outreach membership	National security/law enforcement agency	OBS-1, OBS-5	May exist individual memberships vary	Ensure key security and management personnel hold membership establish a facility-level relationship with regional agency contacts.
Regulatory authority cybersecurity notification relationship	National nuclear regulatory authority	OBS-1, OBS-4, OBS-5	Exists as a regulatory requirement in most jurisdictions	Verify bidirectional: confirm facility is receiving regulatory-authority-originated threat notifications, not only reporting outbound
Protected critical infrastructure information program access (national equivalent)	National homeland security authority	OBS-4, OBS-5	May exist requires application in most jurisdictions	Apply for access if not already authorized designate authorized personnel for DBA-MA decision roles
Facility security	National security	OBS-4, OBS-5	Exists for some	Ensure all DBA-MA

clearances (applicable national level)	sponsoring agency		personnel at nuclear facilities	Tier activation decision-chain personnel hold required clearances initiate upgrades as needed.
Classified briefing access through regulatory or security agency channels	National security agencies, regulatory authority	OBS-5	Ad hoc not proceduralized for DBA-MA triggers in any known implementation	Formalize request for inclusion in sector-specific classified briefing distribution establish classified briefing facility access.
Energy sector risk management agency threat briefing access	National energy sector risk management agency	OBS-2, OBS-5	May exist for some utilities varies by jurisdiction	Establish a relationship with the applicable agency and request inclusion in the energy-sector threat-briefing distribution
Nuclear Facility Military Threat Coordination Protocol	National defense, intelligence, regulatory, and security agencies (proposed)	OBS-3, OBS-4, OBS-5	Does not exist in any national implementation	Advocate for establishment through industry channels in the applicable jurisdiction. Requires interagency agreement or equivalent governmental instrument.
Commercial threat intelligence service	Vendor selected by the facility	OBS-1, OBS-2	Varies by facility	Evaluate and procure an OT-capable threat intelligence service with energy sector specialization ensure integration with SOC/NOC monitoring.
Geopolitical risk monitoring service	Vendor selected by the facility	OBS-3	Uncommon in nuclear utility operations	Evaluate need for structured geopolitical risk monitoring may be addressed at the corporate rather than the facility level.

5. Implementation Recommendations

5.1 Immediate Actions (No Regulatory Change Required)

The following actions can be taken by any AOO immediately, using existing authorities, programs, and commercial services. Verify active participation in the ISAC for the electric utility sector and confirm receipt of all relevant product lines, including any restricted-distribution advisories. Enroll in the national cybersecurity authority's automated indicator sharing program and ensure integration with SOC/SIEM systems. Verify that the regulatory authority cybersecurity notification relationship is functioning bidirectionally. Ensure key DBA-MA decision-chain personnel hold appropriate memberships in national security agency critical infrastructure outreach programs. Establish or verify the relationship with the national energy-sector risk management agency and request inclusion in the energy-sector threat-briefing distribution. Evaluate and procure commercial OT threat intelligence services with energy sector specialization. Designate internal responsibility for geopolitical risk monitoring and DBA-MA trigger assessment.

5.2 Near-Term Actions (Require Coordination with Government Agencies)

The following actions require engagement with specific government agencies but do not require new legislation or regulation. Request formal inclusion in the national security agency classified threat briefing distribution for the facility. Apply for access to the national protected critical infrastructure program if not already authorized. Establish or verify access to a classified briefing facility for receiving classified briefings. Engage the national regulatory authority to formalize the bidirectional threat notification pathway. Initiate security clearance actions for any DBA-MA decision-chain personnel who do not currently hold active clearances.

5.3 Industry-Level Actions (Require Collective Advocacy)

The following actions require collective industry effort through organizations such as the national nuclear industry association, the governing body of the electricity sector ISAC, and the nuclear sector coordinating council or its national equivalent. Advocate for the establishment of the Nuclear Facility Military Threat Coordination Protocol through appropriate policy channels in the applicable jurisdiction. Engage with national regulatory, defense, and security agencies in developing a unified Tier 3 decision framework. Support the development of an extended liability protection framework for DBA-MA preemptive actions. Participate in the development of DBA-MA exercise requirements that test the full information pathway from intelligence source to plant operator.

5.4 Standing Monitoring Posture

Regardless of the current threat environment, the AOO should maintain a standing monitoring posture to ensure information pathways function as needed. This includes regular review (at least quarterly) of all data-sharing agreements and relationships to confirm they remain active and current; periodic testing of communication channels, including backup channels, to verify they

function under degraded conditions; participation in sector-wide exercises that test the flow of threat information from government sources to facility operators; and maintenance of a documented baseline of normal threat reporting volume and pattern against which changes can be recognized.

[Note: This guidance will be updated as the proposed Nuclear Facility Military Threat Coordination Protocol is developed in each national implementation and as national regulatory frameworks for the DBA-MA are formalized. The immediate and near-term actions identified above should not be deferred pending those developments; they represent prudent preparedness measures that strengthen the facility's information posture regardless of the ultimate regulatory outcome.]

5.5 UAS Threat Characterization Review Cycle

DBA-UxS, Section 9, establishes a mandatory review cycle for the threat characterization of uncrewed systems: a full reassessment at intervals not exceeding two years for facilities in a high-threat-profile environment and three years otherwise, and an unscheduled reassessment within ninety days of any of four intelligence triggers, with notification of the regulatory authority within thirty days. DBA-UxS governs the intervals and the triggers. This section specifies the operational mechanics by which a nuclear facility meets them and assesses the triggers in coordination with national intelligence authorities.

5.5.1 Periodic Review

Each facility shall conduct a comprehensive review of its threat characterization for uncrewed systems at the interval DBA-UxS, Section 9.1, sets for its threat profile: not exceeding two years in a high-threat-profile environment and three years otherwise. The threat profile is determined through the national coordination mechanisms, not by the facility alone, and a nuclear facility in a geopolitically elevated environment should expect the two-year interval. The review shall address uncrewed aerial, surface and undersea systems even where the assessed threat is concentrated in one domain. It shall assess whether the existing characterization remains adequate in light of developments in adversary capability, counter-UAS technology, autonomous systems, and relevant conflict experience accumulated during the review period, drawing on the sources identified in Sections 3 and 4 of this document and on any classified threat assessments made available through the coordination mechanisms described in the applicable national implementation.

The periodic review shall produce a documented determination of whether the existing UAS threat characterization remains adequate or requires revision. Where revision is indicated, the facility shall update the threat characterization and assess the implications for the engineering and operational provisions established under DBA-UxS and DBA-MA-UxS-NA, including passive kinetic defense specifications, concealment measures, counter-UAS capability

requirements, and supply chain provisions. The review and its findings shall be reported to the national regulatory authority.

5.5.2 Intelligence Trigger Assessment Procedures

DBA-UxS, Section 9.2, defines four intelligence triggers, any one of which shall initiate an unscheduled reassessment within ninety days. The following procedures specify how the facility monitors for and assesses each trigger using the monitoring pathways of this document.

Trigger A: confirmed acquisition or deployment by a relevant adversary of a new uncrewed platform class, variant or capability, including AI-enabled autonomous operation, improved warhead mass, improved range or improved countermeasure resistance, that moves the threat class above the design-basis class used in the facility's current hardening assessment. The facility shall monitor for such developments through the intelligence reporting channels of OBS-5, the open-source conflict observation of OBS-3, and commercial technology and threat intelligence services. The facility's technical review committee shall determine promptly whether a reported development moves the threat class above the design-basis class, and a positive determination starts the ninety-day period.

Trigger B: demonstrated operational employment of uncrewed systems by a relevant adversary against energy, water, data center or comparable critical infrastructure in any theater of operations. The facility shall monitor for such employment through open-source conflict reporting under OBS-3 and intelligence reporting under OBS-5. The threshold is verified employment against comparable infrastructure; it does not require an attack on nuclear facilities, and employment of AI-enabled systems in any active conflict should be assessed under this trigger and Trigger A together.

Trigger C: credible specific threat reporting, from classified or unclassified sources, of an adversary's intention to conduct operations with uncrewed systems against the facility, its facility category, or infrastructure in the host nation or allied nations. Such reporting will usually reach the facility through the regulatory authority relay described for OBS-4 or the classified briefing pathways described for OBS-5, and the facility shall ensure that those pathways are structured to carry it.

Trigger D: material deterioration in the geopolitical environment, including significant deterioration of relations with a state capable of employing uncrewed systems, mobilization or forward deployment of such forces, or a declared threat condition affecting the facility's environment, that materially changes the assessed probability that the design-basis threat will be realized. This trigger is monitored through OBS-3 and OBS-4 and overlaps closely with the Tier 1 and Tier 2 indicators of DBA-MA L3. Activation of either tier on these grounds shall be treated as initiating the unscheduled reassessment.

5.5.3 Coordination with National Intelligence Authorities

The trigger assessment procedures require information that is not fully available through unclassified channels and commercial services. Classified intelligence assessments of adversary UAS capability development, acquisition programs, and proliferation activity are essential inputs to Triggers A, B and C. The facility's participation in the intelligence coordination mechanisms described in Section 4 of this document provides the primary pathway for receiving this information. Where the facility's own assessment identifies a potential trigger condition through unclassified sources, the facility shall request a classified assessment through the appropriate coordination channel to confirm the development's materiality before concluding that a trigger has been met.

The periodic review shall include, as a standing element, a request to the national intelligence authorities for a current assessment of adversary UAS capabilities relevant to the nuclear facility threat profile. This request should be coordinated through the applicable national regulatory authority or, when established, through the Nuclear Facility Military Threat Coordination Protocol.

5.6 Phase 3 ASSC Maintenance Monitoring: Passive Kinetic Defense Installation Integrity

DBA-UxS (Section 8.2) and DBA-MA-UxS-NA establish passive kinetic defense barrier installations as a credited protection layer for systems, structures, and components that cannot be practicably hardened, buried, or concealed but remain credited in the ASSC safety analysis. Unlike hardened structures or below-grade installations, passive kinetic defense barriers are sacrificial by design: their protective function is performed through destruction upon contact with an incoming munition. The consequence is that a strike event that engages a barrier consumes the barrier. Until the affected barrier section is restored, the protected asset reverts to unshielded exposure. Phase 3 ASSC maintenance monitoring must therefore include passive kinetic defense installation integrity as a tracked parameter, with post-strike inspection and restoration as defined operational obligations.

5.6.1 Post-Strike Inspection Requirement

Following any UAS strike event at or near the facility, the AOO shall conduct a physical inspection of all passive kinetic defense installations to assess their post-strike condition. The inspection shall identify barrier sections that have been destroyed, damaged, displaced, or degraded below their design-basis configuration. Inspection results shall be documented against the design-basis barrier layout established under DBA-UxS, Section 8.2, recording the location, extent, and nature of any deficit. For each deficit, the documentation shall identify the specific credited ASSC component or group of components whose passive kinetic defense coverage is reduced or absent, and shall assess the residual protection provided by the remaining defense-in-depth layers for those assets pending barrier restoration. The inspection shall be completed within the shortest practicable interval consistent with personnel safety under the prevailing threat conditions.

5.6.2 Restoration Confirmation and Documentation

Restoration of degraded or destroyed barrier sections shall be conducted using the on-site inventory of barrier replacement materials maintained under DBA-MA L2, Section 4.2.6 (mesh, cable, structural hardware, and fasteners). Upon completion of restoration activities, the AOO shall confirm that each restored barrier section meets the design-basis configuration specified in the facility's passive kinetic defense design under DBA-UxS, Section 8.2, including mesh parameters, standoff distance, and structural integrity. Confirmation shall be documented in the Phase 3 ASSC maintenance log as a restoration record, with the date, scope of restoration, and confirmation of design-basis compliance. Until restoration is confirmed, the affected assets shall be carried in the ASSC maintenance monitoring record as operating with reduced passive kinetic defense coverage, and the Phase 3 contingency procedures shall address the compensatory measures available for those assets under the remaining defense-in-depth layers.

6. References

6.1 DBA-MA Series Documents

DBA-MA International Framework (L1) — Expanding the Design Basis for Nuclear Power Plants to Address Military Action Against Nuclear Facilities: An International Framework. Eclectic Technologies.

DBA-MA L2 — Design Basis Accident: Military Action Against Nuclear Facilities. Formal Accident Description and Design Requirements. Eclectic Technologies.

DBA-MA L3 — Trigger Framework for Military Action Design Basis Accident: Observable Indicators and Decision Criteria for Transition to Autonomous Safe Shutdown Condition. Eclectic Technologies.

DBA-UxS — Uncrewed Systems: Threat Considerations for Critical Infrastructure. Eclectic Technologies.

DBA-MA-UxS-NA — Uncrewed Systems: Nuclear Sector Implementation Annex. Eclectic Technologies.

6.2 IAEA References

IAEA Safety Fundamentals No. SF-1 — Fundamental Safety Principles (2006).

IAEA Safety Standards Series No. SSR-2/1 — Safety of Nuclear Power Plants: Design (Rev. 1, 2016).

IAEA Nuclear Security Series No. 10-G (Rev. 1) — National Nuclear Security Threat Assessment, Design Basis Threats and Representative Threat Statements (2021).

IAEA Nuclear Security Series No. 42-G — Computer Security for Nuclear Security (2021).

IAEA, Nuclear Safety, Security and Safeguards in Ukraine — Director General summary reports and updates (2022–present).

6.3 National Implementation References

National regulatory and legislative references applicable to each implementation of this document are identified in the corresponding national child document. The United States national references are set out in DBA-MA L4, United States National Implementation.

Revision History

Version	Date	Author	Description
2.0	April 2026	Tim Roxey	Elevation to IAEA-level instrument; US national implementation content placed in Annex A; jurisdiction-neutral framework established
2.1	April 2026	Tim Roxey	Annex A — United States National Implementation extracted to DBA-MA L4, United States National Implementation (v1.0); inline Annex A references updated to national child document citations; L1 version reference updated to v1.4
2.2	April 2026	Tim Roxey	IAEA L1 parent reference updated from v1.3 to v1.4 throughout the body and revision history.
2.3	April 2026	Tim Roxey	Section 5.5 UAS review cycle references updated throughout: DBA-MA-UAS addendum replaced with DBA-UxS (v1.0) and DBA-MA-UxS-NA (v1.0) pair. DBA-MA-UAS Section 4.6.7 references updated to DBA-UxS Section 4.6.7. Bibliography entry updated to DBA-UxS and DBA-MA-UxS-NA.
3.0	September 2026	Timothy E. Roxey	First final version, aligned with the DBA-MA

			International Framework, Version 2.0. Removes the draft status and adds the citation block, stamped with the reserved version DOI 10.5281/zenodo.22945520. Restores the agreement and relationship readiness table in Section 4, which versions 2.2 and 2.3 had replaced in error with the revision history table; the restored table is that of version 2.1, with “National Security Agency” corrected to the generic “national security agency”. Moves the revision history to the end. Aligns Section 5.5 with DBA-UxS, Section 9: two-year or three-year intervals by threat profile, all three uncrewed domains, and the four intelligence triggers A to D as DBA-UxS defines them, each mapped to this document’s monitoring pathways. Corrects the DBA-UxS barrier references from Sections 4.6 and 4.6.7 to Section 8.2. Makes the industry organizations in Section 5.3 and the energy sector agency in the Section 2 table generic. Adds the drone-detection dependency to Section 1. Cites series instruments by designation without version numbers and names the Nuclear Security Series documents relied on.
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